

6.5 Adding Integers

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.		 Learning Target	I can add integers with procedural fluency.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.		MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- What are additive inverses? - How can you use the signs of addends and their absolute values to determine the sign of their sum? - Describe how to add integers with the same signs and with different signs?			
 Lesson Narrative	In this lesson, students begin by modeling additive inverses on number lines and with integer chips. Students discover why the sign of the sum stays the same when the signs of the addends are the same. They also discover why, when the signs of the addends are not the same, the answer is the amount remaining after canceling the zero pairs with the sign of the addend with the greater absolute value.			
 Component Summary	6.5.1 Warm-Up (~5 min.)	Write true addition equations using integers from -3 to 3 (SE p. 255)	6.5.4 Collaboration (10 min.)	Fill in missing addends and sums in integer addition equations (SE p. 257)
	6.5.2 Exploration (10 min.)	Explore the concept of additive inverses (SE p. 255)	6.5.5 Your Turn! (10 min.)	Practice finding the sum of one-, two-, and three-digit integers (SE p. 258)
	6.5.3 Guided Instruction (10 min.)	Connect additive inverses to zero pairs for adding with the same or opposite signs (SE p. 256)	6.5.6 Wrap-Up (~5 min.)	Reflect on what they are still unsure about when adding integers (SE p. 258)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Existence of additive inverses <u>Additional Terms:</u> - Absolute value - Integer		Integer chips	Consider creating an anchor chart to illustrate how to add integers with the same and with different signs. Include instructions, visual models (number line and integer chips), and a few examples.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to explain how many zero pairs are created in 6.5.2 Guided Instruction. - Students may have difficulty finding the missing addends in 6.5.4 Collaboration. - Students may add integers of opposite signs and give the value a negative sign. For example, $-3 + 2 \neq -5$.	- Consider displaying an anchor chart that illustrates how to add integers with the same and different signs. Include instructions, visual models (number line and integer chips), and a few examples. - Consider rewriting expressions in 6.5.3 Guided Instruction to show the additive inverses involved as students develop procedural fluency when adding without manipulatives or number lines. - Consider allowing students to use the models that make sense to them to add integers if they are not yet at a place to do so without. Encourage the use of number-line models in this case because of the size of the addends in the lesson.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns	MA.K12.MTR.4.1: Discussion and Reflection
	In 6.5.4 Collaboration, consider having students take turns explaining how they added the integers with different signs. These discussions will improve student understanding and confidence.	6.5.4 Collaboration provides an opportunity for students to engage in mathematical discussion through explaining their mathematical process and justifying their results to a peer. Use this as an opportunity for students to practice articulating their mathematical thinking and comparing their methods to others.
Each activity in this lesson provides opportunity to differentiate instruction by content, by process, or by product. - Provide integer chips for 6.5.2 Exploration and 6.5.4 Collaboration for kinesthetic and visual entry points. - Offer verbal responses for students to submit via phones or apps for the 6.5.6 Wrap-Up. - For 6.5.5 Your Turn!, consider substituting the given integer addition problems with less difficult problems if students are struggling. If students finish early, supplement with more complex problems, such as four-digit integers or sums with three addends.		
Lesson Notes:		

6.5.1 Warm-Up: True Equations

Component Overview: Students fill in the blanks to write two addition equations using integers from -3 to 3 .



6.5.1 Warm-Up: True Equations

- Using integers between -3 and 3 one time each, fill in the blanks to make the equation true.

Answers will vary.

$$\boxed{-3} + \boxed{2} = \boxed{-1}$$



For students who may finish early, consider having them create as many true equations as they can.



MA.K12.MTR.3.1: Fluency

*Did you fill in the addends or the sum first?
Which method made more sense for you? Why?*

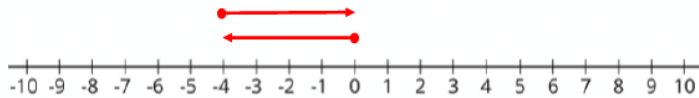
6.5.2 Exploration: Additive Inverses

Component Overview: Students explore multiple representations of additive inverses and finding their sum. Students see additive inverses represented on number lines and with integer chips. Students connect additive inverses to zero pairs. Finally, students create an example of an additive inverse and share their example with a partner.



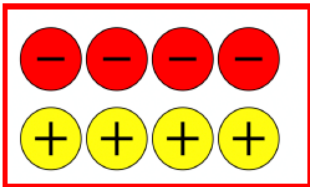
6.5.2 Exploration: Additive Inverses

1. Use the number line to show the sum of $-4 + 4$. Then, complete the equation.

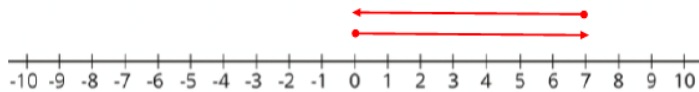


$$-4 + 4 = 0$$

2. Represent the sum of $-4 + 4$ using integer chips.

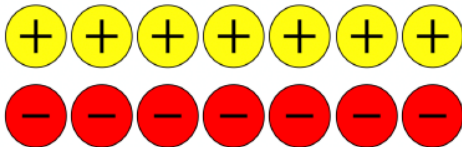


3. Use the number line to show the sum of $7 + (-7)$. Then, complete the equation.



$$7 + (-7) = 0$$

4. Represent the sum of $7 + (-7)$ using integer chips.



Consider using question stems for students to answer on their own or for you to ask while circulating.

- What do you notice about the sum of each question?
- What do you notice about the values being added?
- Do additive inverses remind you of anything you learned about when modeling integers with integer chips?



Existence of Additive Inverses

For every a there exists $(-a)$ so that $a + (-a) = 0$ and $(-a) + a = 0$.

6.5.2 Exploration: Additive Inverses (continued)

Additive inverses are another way to represent the concept of a zero pair.

5. What number is the additive inverse of -15 ?

15

6. What number is the additive inverse of 27 ?

-27

7. Choose any positive integer and write it in the blank:

Answers will vary. Any positive integer is correct. The example provided here uses 12.

Use your number to fill in the blanks in the statement.

For the value 12 , there exists (-12) so that

$12 + (-12) = 0$ and $(-12) + 12 = 0$.

8. Write your partner's positive integer:

Answers will vary. The example provided here uses 9.

Use your partner's number to fill in the blanks in the statement.

For the value 9 , there exists (-9) so that

$9 + (-9) = 0$ and $(-9) + 9 = 0$.



How do you know that 15 is the additive inverse of -15 ?

6.5.3 Guided Instruction: Adding Integers

Component Overview: Students summarize concepts explored in previous lessons adding integers and relate these concepts to how to think about adding integers without using manipulatives or number-line models. They formalize this learning as they continue developing procedural fluency.



6.5.3 Guided Instruction: Adding Integers

Understanding the existence of additive inverses can be helpful when adding integers with larger absolute values.

1. Complete each statement based on your understanding of adding integers so far.

- a. When adding two positive integers, the sum is

- ☐ positive.
- ☐ negative.

- b. When adding two negative integers, the sum is

- ☐ positive.
- ☐ negative.

When adding a positive and a negative integer, the sum will be either positive or negative, depending on which value has the larger absolute value.

2. Consider the expression $40 + (-165)$.

- a. What will be the sign of the sum?

The sign of the sum will be negative.

- ☐ positive
- ☐ negative

- b. Explain your reasoning.

The sum will be negative because -165 has the greater absolute value.



How does the number of zero pairs formed relate to additive inverses?



Consider rewriting the expression as $40 + (-40) + (-125)$ and think aloud to compare the two expressions and discuss how this shows how the additive inverse of 40 can be used to consider the zero pairs created and eliminated when adding $40 + (-165)$. The chips that would remain after eliminating the zero pairs are 125 negative chips, or -125 .

6.5.3 Guided Instruction: Adding Integers (continued)

When adding these integers, zero pairs are created.

- c. Use additive inverses to explain how many zero pairs are created.

40 zero pairs are created. There are the 40 positive integers, from the addend 40, and 40 negative integers, from the -165 addend, that can be combined to form zero pairs.

- d. What remains after combining the additive inverses?

125 negative integers remain.

- e. Complete the equation.

$$40 + (-165) = -125$$

3. Complete the statements to describe how to find the sum of $-1,029 + (-38)$.

When adding $-1,029 + (-38)$, ☐ zero pairs
☒ no zero pairs are created because the signs of the addends are

- ☒ the same.
☐ opposites.

The sum can be found by adding the absolute values of the addends. Since both addends are

- ☐ positive,
☒ negative,

the ray for the second addend of the sum will go farther to the ☐ right ($\rightarrow$)
☒ left ($\leftarrow$) of zero on a number

line.

The sum of $-1,029 + (-38)$ is

- ☐ 991.
☐ -991.
☐ 1067.
☒ -1067.



If students struggle to understand why 40 zero pairs are created, consider modeling a problem with smaller values to model how the zero pairs are created using integer chips.



Utilize talk moves to position students for synthesis. "Karma, in your own words, how would you describe the way to use zero pairs for adding integers? Chantoy, restate Karma's approach but in your own words."

6.5.4 Collaboration: Reasoning with Integers

Component Overview: Working with a partner, students determine the missing addends in integer addition equations. Students finish the activity by explaining how they determined the missing addends.



6.5.4 Collaboration: Reasoning with Integers

1. Work with your partner to fill in the blanks with an integer to make each equation true.

$$\boxed{14} + \boxed{-49} = \boxed{-35}$$

$$\boxed{304} + \boxed{-70} = \boxed{234}$$

$$\boxed{-136} + \boxed{161} = \boxed{25}$$

2. Choose one equation and explain your thinking in determining the missing value.

Student choice will vary.

$$\boxed{14} + \boxed{-49} = \boxed{-35}$$

Since 14 is positive and the sum is negative, the other addend must be negative. The sign of the sum indicates which addend is the integer with the greater absolute value. Therefore, the missing addend will be negative.

To determine the missing value, I added the absolute values of both existing numbers in the equation, so that zero pairs can be created. Then, I kept the sign of the sum. $|14| + |-35| = 49$. The missing addend is therefore -49 .



Turn and Talk

This routine provides students an opportunity to share their thoughts, reflect, and finalize their answers. To increase active listening, consider asking partners to explain each other's work.

6.5.5 Your Turn!

Component Overview: Students continue practice finding the sum of one-, two-, and three-digit integers.



6.5.5 Your Turn!

1. Find each sum.

a. $10 + (-204) = -194$ b. $(-99) + (99) = 0$

c. $904 + (-11) = 893$ d. $-1 + 76 = 75$

e. $205 + (-643) = -438$ f. $-151 + (-70) = -221$

g. $-399 + (-405) = -804$ h. $-80 + 606 = 526$

i. $703 + (-190) = 513$ j. $-600 + (-78) = -678$



- How can you determine the sign of the sum when the integers have the same sign?
- How can you determine the sign of the sum when the integers have opposite signs?



Consider displaying an anchor chart with instructions, visual models, and examples for adding integers with the same and with opposite signs. Referencing this chart will help students develop procedural fluency with adding integers.

6.5.6 Wrap-Up: Growth Mindset

Component Overview: Students share what they are still unsure about and develop a plan for how they can fill in the gap(s). Consider using students' perceptions of their understanding, along with their accuracy on the questions in 6.5.5 Your Turn!, to determine which students need support before the next lesson.



6.5.6 Wrap-Up: Growth Mindset

1. Today, I am still unsure about _____

_____.

2. To fill in this gap, I intend to _____

_____.

Answers will vary.



Consider providing answer frames for students to select all that apply.

- The meaning of additive inverse
- The meaning of absolute value
- How to use integer chips to represent zero pairs
- Fluency adding integers with the same sign
- Fluency adding integers with opposite signs